

Sindy
Software Engineer | Generative AI & Python

AI engineer with hands-on experience building production-grade RAG systems, LLM applications, and data pipelines. Proven track record developing enterprise NLP solutions including structured query extraction, hybrid retrieval architectures, and LLM fine-tuning. Strong foundation in RAG, graph and vector databases, embedding strategies, and end-to-end ML system deployment.

EXPERIENCE

Aurevia Capital **Jun 2025 – Present**
Unified data intelligence platform for collectibles pricing and market analytics

Co-Founder

- Designed and implemented end-to-end ML system including automated web scraping, feature engineering, model training, and inference APIs for a luxury watch price forecasting platform
- Collaborated directly with prospective clients and investors to translate business requirements into technical solutions, refining product strategy through iterative feedback
- Accepted into INSEAD Venture Lab's AI Founder Sprint 2025 incubator

AI Apprenticeship Programme (AIAP) | AI Singapore **Sep 2024 – May 2025**

Associate AI Engineer

LLM-Powered HR Query System (Industry Collaboration)

Collaborated with global HR analytics leader to develop LLM-powered structured query system for enterprise HR data

- Built end-to-end pipeline to extract structured HR metadata (intents, metrics, filters) from natural language queries
- Designed custom DSPy optimization workflows integrated with Langfuse observability for real-time model performance monitoring and prompt engineering iteration
- Deployed multi-strategy inference architecture (full prompt, query decomposition, multi-hop) with InstructorAI
- Implemented custom evaluation framework with interpretable metrics including cosine similarity, Jaccard, and set overlap across 7 metadata categories
- Fine-tuned BERT and open-source LLM models with Unsloth and QLoRA for memory-efficient parameter updates

LLM-Powered Knowledge Assistant (Capstone Project)

- Developed a containerised RAG system for multi-format document querying using LLaMA 3.1 Instruct, Streamlit frontend, and Ollama backend infrastructure with Kubernetes orchestration for scalable deployment
- Engineered hybrid embedding strategy using MiniLM (text) and GraphCodeBERT (code) with FAISS vector indexing

Gospell Digital Technology Co. – Shenzhen, China **Feb 2018 – Jan 2023**
Digital technology company specialising in digital broadcasting, satellite communication, and smart systems

Senior Financial Accountant

- Conducted rigorous financial data analysis and validation for digital broadcasting products, advising leadership on resource allocation and budgeting strategies while ensuring data integrity across complex financial records

EDUCATION

Singapore Management University **Aug 2023 – Mar 2025**
Master of IT in Business (Artificial Intelligence track)

Relevant Coursework: Natural Language Processing for Smart Assistants (A), Introduction to Artificial Intelligence (A), Data Management (A), Applied Machine Learning (A-), Deep Learning for Visual Recognition (A-), AI Planning and Decision Making (A-)

Teaching Assistant: COR1305 Spreadsheet Modelling and Analytics; ISFS626 Web3 in Digitalised Currencies and CBDCs

Nanyang Technological University **Jul 2014 – Jun 2017**
Bachelor of Accountancy

TECHNOLOGIES

ML/AI: LangChain, RAG systems, LLM fine-tuning (QLoRA, Unsloth), DSPy, vector databases (FAISS), PyTorch

Infrastructure: Docker, Kubernetes, AWS

Frontend: React, TypeScript, Tailwind CSS, shadcn/ui

Backend & Data: Python, FastAPI, PostgreSQL, Supabase, SQL

PROJECTS

Kube – Crypto Sentiment Analyzer

Feb 2026

- Built real-time cryptocurrency sentiment pipeline on Kubernetes, scraping across 9 Reddit forums every 5 minutes to track social sentiment alongside price data for BTC, ETH, SOL, BNB, and XRP
- Designed event-driven microservices architecture using NATS JetStream for async messaging, decoupling ingestion (CronJob), analysis (Deployment), and persistence (Deployment) into independently deployable Python services
- Implemented core Kubernetes patterns: CronJobs for scheduled scraping, StatefulSets with PVCs for PostgreSQL persistence, InitContainers for dependency ordering, and Helm for third-party chart management

Adversarial AI Security Detection

Nov 2024

Research & Implementation of AI Security Techniques

- Trained a backdoor model using WaNet to create a simulated adversarial ML environment
- Implemented detection techniques (Neural Cleanse, ABS, and STRIP) to identify if given set of CNN models is malicious
- Reverse-engineered malicious triggers in neural models and improved model robustness through defence strategies
- Applied AI security principles and interpretability tools to enhance trust in machine learning deployments

Food Social Media Platform with ML Classification

Jun 2024

- Built a specialised social media platform for food enthusiasts with ML-powered image classification using a subset of FoodSG-233 dataset (120K+ images, 100 categories)
- Implemented transfer learning with lightweight models (EfficientNet, MobileNet, SqueezeNet) for mobile-optimised food recognition, incorporating Label Studio for user-guided annotation refinement
- Designed AWS-based architecture with automated ETL pipelines, A/B testing deployment, and active learning to handle model drift and evolving food trends